A) Select the most suitable answer for the following statements: To be answered in (15) Minutes [ ] / 2 Marks

Question	1	2	3	4	5	6
Answer						

[1.5 Marks]

1- **Which of the following network devices is considered a host?**
   A) Router
   B) Switch
   C) Printer
   D) Wi-Fi Access point

2- **In FDM, multiple channels are transmitted by?**
   A) Sharing time slots.
   B) Sending packets sequentially.
   C) Using different frequency bands simultaneously.
   D) Using different routers.

3- **Circuit switching provides:**
   A) Guaranteed performance.
   B) Shared resources.
   C) Packet loss.
   D) All of the above are correct.

4- **Which layer of the TCP/IP stack is responsible for data transfer between neighboring network elements (directly connected nodes)?**
   A) Application layer. B) Transport layer. C) Network layer
   D) Link layer. E) Physical Layer

5- **Which of the following terms refers to a chunk of data with a header at the Network layer?**
   A) Message B) Segment C) Datagram
   D) Frame E) packet

6- **The applications of file transfer over internet requires one of the following to be more effective:**
   A) Low delay
   B) 100% of data integrity.
   C) low value of throughput
   D) None of the above

B) Define the Protocol in the context of computer networks. [0.5 Mark]

Question 2:

To be answered in (20) Minutes [ ] / 7 Marks

Part#1 : Select the most suitable answer for the following statements: [2 marks]

Question	1	2	3	4	5	6	7	8
Answer								

1- Which of the following HTTP request methods is used by a client to retrieve data from a server by including data as query parameters appended to the URL?

A) Post Method B) Get Method

C) Head Method D) Put Method

2- Which of the following HTTP status codes indicate that the server could not understand the request message sent by the client?

A) 300 B) 400

C) 404 D) 505

3- Which of the following statements about a TCP socket is correct?

A) A TCP socket is identified by a 4-tuple: source IP address, destination IP address, source port number, and destination port number.

B) Only destination port number used for demultiplexing.

C) Server supports only one TCP socket at a time.

D) All the above are correct.

4- Which of the following is advantage of UDP protocol?

A) No setup/handshaking needed (no RTT incurred)

B) Can function when network service is compromised.

C) It includes a checksum mechanism for basic error detection.

D) All of the above are correct

5- Using TCP fast retransmit is performed _____.

A) If sender receives 3 ACKs for the same data

B) If sender receives 3 duplicate ACKs for the same data

C) If sender is unsure about whether the receiver has received a packet

D) None of the above

6- Which of the following TCP header flags is set to 1 in the **first message** sent during TCP connection establishment?

A) FIN B) SYN

C) RST D) ACK

7- TCP provides a flow-control service to:

A) Eliminate network congestion B) Eliminate overflowing receiver's buffer

C) Eliminate overflowing sender's buffer D) All the answers are correct.

8- Which of the following of data transfer protocols has the best performance in term of utilization

A) Stop and wait protocol B) Go Back-N protocol

C) Selective repeat protocol D) All the answer has the same performance term of utilization.

Part#2:

1- **Write the type of DNS record in the Blank:** **[1 Mark]**

(networkutopia.com, dns1.networkutopia.com, ……..)

(dns1.networkutopia.com, 212.212.212.1, ……..)

2- **Suppose the HTTP protocol uses a conditional GET request with the header: (*If-Modified-Since: <date>*), If the resource on the server has been updated after this date, what will the server response be? Will the data be downloaded, and what is the status code in the response?** **[1 Mark]**

Response:

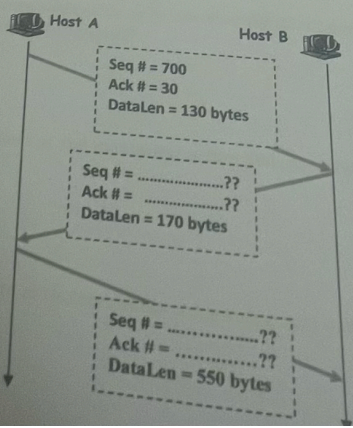
Status code:

3- **Why is web caching used? List two reasons.** **[1 Mark]**

1-

2-

4- **Suppose** Hosts A and B are using TCP as shown in the figure below. Complete the missing Seq/Ack numbers. ***[2 mark]***

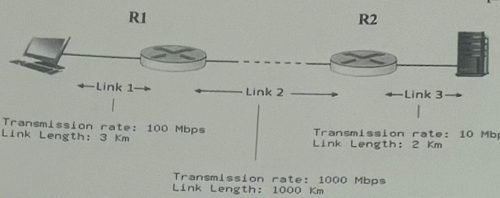


Question 3:

To be answered in (25) Minutes [ ] / 5 Marks

Write a detailed answer of the following questions, write the unite used if needed:

A) Consider the figure below, with three links, each with the specified transmission rate and link length. Assume the length of a packet is 500 **Megabyte's**. The speed of light propagation delay on each link is 3x10^8 m/sec. **[2.5 Marks]**



Questions

1. **What is the transmission delay of link 1?**

2. **Compute the propagation delay at Link 1.**

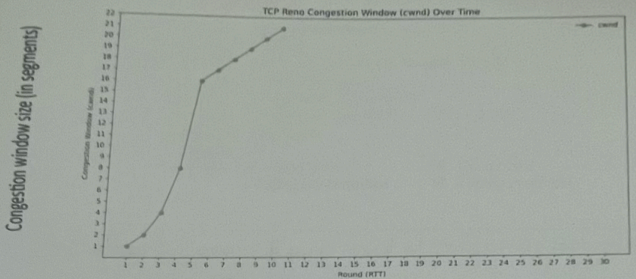
3. **What is the maximum achievable data rate that the client can send to the server over all three links in Mbps (Throughput)?**

4. **If the client sends data at the maximum achievable data rate, what is the utilization of Link 3 (Utilization)?**

5. **If the client sends data at the maximum achievable data rate, what is the traffic intensity at router (R1)?**

Student Name (in English): _____ Student ID: _____

B) Consider the figure below, Assuming TCP Reno is the protocol experiencing the behavior shown, answer the following questions. *[2.5 marks]*



1. **Identify the intervals of time when TCP slow start is operating.** **[0.5 Mark]**

2. **Identify the intervals of time when TCP congestion avoidance is operating.** **[0.5 Mark]**

3. **At the 10th transmission round, supposed the segment loss is detected by time out, then compute the new value of Congestion Windows (cnwd) at RTT=11?** **[0.5 Mark]**

4. **Consider the question 3 above, what is the TCP congestion approach operating at RTT 16, suppose no more loss detected? (Slow start or congestion avoidance or fast recovery) and why?** **[0.5 Mark]**

5. **What is the value of cnwd at transmission round (RTT = 16) ?** **[0.5 Mark]**